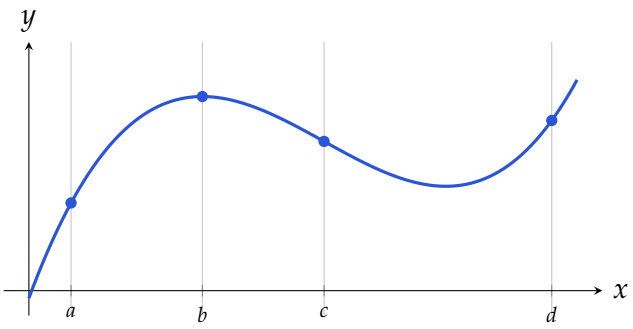
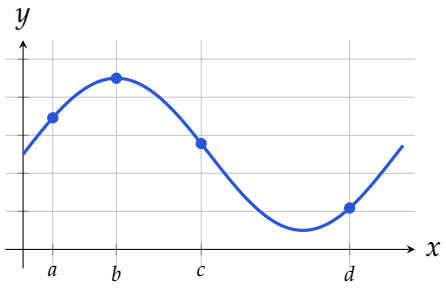
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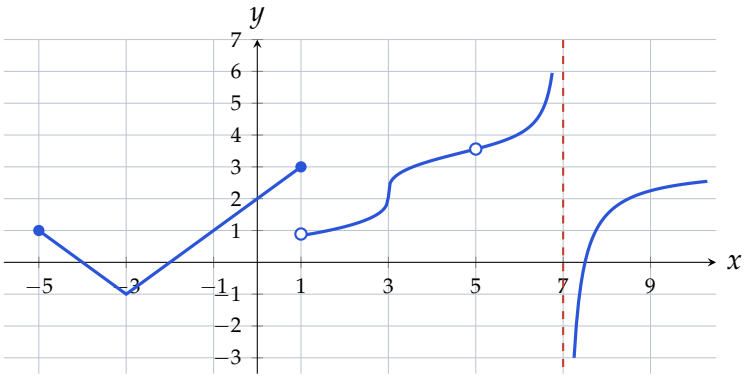
Unit 1 • Lesson 1-7 • Guided notes

Lesson 1-7 • Instantaneous Rates of Change

Guided notes • Fill in the blanks and work the examples in the boxes

Objective: draw tangent lines, read the sign of the derivative, and recognize where a graph is not differentiable.

Rates on a graph	 The instantaneous rate of change at a point is the slope of the _____: the line that just touches the graph at that point.
Synonyms	Instantaneous rate of change = _____ = _____ = _____ = _____
Example 1	Sketch the tangent line at each point and decide whether the derivative is positive, negative or zero.  $g'(a)$ _____ $g'(b)$ _____ $g'(c)$ _____ $g'(d)$ _____
Differentiability	A function is **not differentiable** (f' does not exist) at: 1) _____ 2) _____ 3) _____

Example 2	Find the x-coordinates where h is not differentiable, and say why. 
Summary	